

Safety Data Sheet

According to Notification of Thailand's Ministry of Industry B.E. 2555 (2012)



HYDRAULIC AW 46

Version 0.0

Release Date: 2018/01/30

1. PRODUCT AND COMPANY IDENTIFICATION

Product name : HYDRAULIC AW 46
Other means of identification : Lubricating oil
Recommended use : This product should not be used for purposes other than those
Limitation of use : specified without the advice of an expert
PETRONAS International Marketing (Thailand) Co.,Ltd
Importer name : 0105555014294
Company registration number : 388 Exchange Tower, Unit Nos. 4101-4104 41st floor,
Registered corporate address : Sukhumvit Road, Kweang Klongtoey, Khet Klongtoey,
Bangkok 10110
+662-663 5881-5
Emergency phone numbers :

2. HAZARDS IDENTIFICATION

GHS Classification

Skin corrosion / Irritation

Category 3

Serious eye damage / Eye irritation

Category 2B

GHS Label element

Pictogram No

Signal word Warning

Hazard statement(s) H316: Causes mild skin irritation
H320: Caused eye irritation

Precautionary statement(s)

Prevention P261: Avoid breathing vapours
P264: Wash thoroughly after handling
P271: Use only outdoors or in a well-ventilated area
Response P304+P340: If inhaled: Remove victim to fresh air and keep at rest in a position comfortable for breathing
P305+P351+P338: If in eye: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do.
P332+P313: If on skin, wash with plenty of soap and water
Storage Store in a well-ventilated place. Keep container tightly closed

Other hazards which do not result in classification

No information available.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture : Mixture

Hazardous components

Chemical Name	CAS-No.	Concentration (%)
Distillates (petroleum), solvent- dewaxed heavy paraffinic	64742-65-0	95 – 99%
Additive	Proprietary	1 - 5%

4. FIRST AID MEASURES

Description of first aid measures

Inhalation	Remove person to fresh air. If not breathing, give artificial respiration. Get medical attention
Skin contact	Wash contact areas with soap and water. Launder contaminated clothing before reuse
Eye contact	Flush thoroughly with water for at least 15 minutes. If irritation occurs, get medical attention
Ingestion	Do NOT induce vomiting. Get immediate medical attention

Most important symptoms and effects, both acute and delayed: Headache, dizziness, nausea, respiratory irritation, Mildly irritating to skin.

Indication of any immediate medical attention and special treatment needed: Treat symptomatically

5. FIREFIGHTING MEASURES

Extinguishing media

Suitable extinguishing media: Water spray, foam, dry chemical or carbon dioxide (CO₂).

Inappropriate Extinguishing Media: Straight streams of water

Special hazards arising from the substance or mixture

Non- flammable mixtures. Elevated temperatures can lead to the formation of irritating vapors.

Special protective equipment and precautions for fire-fighters

Fire fighters should use self-contained breathing apparatus (SCBA) to fight fires. Use water spray to cool fire exposed surfaces and to protect personnel.

6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures

Evacuate area. Avoid contact with spilled material. Half-face or full-face respirator with filter for organic vapor.

Environmental precautions

Prevent entry into waterways, sewers, basements or confined areas.

Methods and materials for containment and cleaning up

Stop leak if without risk. Move containers from spill area. Absorb with an inert dry material (e.g. sand). And place in waste disposal container.

7. HANDLING AND STORAGE

Precautions for safe handling Avoid breathing vapor or mist. Avoid contact with skin and eyes. Wear suitable gloves, coveralls, apron and boots. Use only with adequate ventilation. Use non-sparking tools. Prevent spills and leaks to avoid slipping hazards.

Conditions for safe storage, including any incompatibilities

Keep containers closed when not in use and check regularly for leaks. Keep in the original container protected from direct sunlight in a dry, cool and well ventilated areas. Store away from incompatible materials. Avoid excessive long-term storage temperatures to prolong shelf life. Maximum storage temperature: 60C

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Control parameters

Mineral Oil	64742-65-0	ACGIH TLV-TWA	5 mg/m ³
		ACGIH TLV-STEL	10 mg/m ³
		OSHA PEL-TWA	5 mg/m ³

Appropriate engineering controls

Ventilation may be used to control or reduce airborne concentrations.

Personal protective equipment

Eye/face protection: Goggle with face shields are recommended.

Skin and Body protection: Wear gloves made from nitrile rubber chemical / oil resistant clothing

Respiratory protection: Wear organic vapor respirator.

Work / Hygienic Practices:

When using do not eat drink or smoke. Wash hand prior to eating ,drinking , smoking or using the toilet. Wash contaminated clothing and other protective equipment before reusing.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance and Colour:	Bright&Clear
Odour	Characteristic
Odour Threshold	No data available
pH	No data available
Melting point/freezing point	No data available
Initial boiling point and boiling range	No data available
Flash point	250 °C
Evaporation rate	No data available
Flammability (solid, gas)	No data available
Upper/lower flammability or explosive limits	No data available
Vapour pressure	No data available
Vapour density	No data available
Relative density	No data available
Water solubility	Insoluble
Partition coefficient: n-octanol/water log Pow	No data available
Auto ignition temperature	No data available

Safety Data Sheet

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Decomposition temperature
Viscosity

No data available
46.48 mm²/s @40 °C

10. STABILITY AND REACTIVITY

Reactivity	No data available
Chemical stability	Stable under recommended storage conditions
Possibility of hazardous reactions	No polymerization
Conditions to avoid	Excessive heat and sources of ignition
Incompatible materials	No data available
Hazardous decomposition products	Does not decompose at ambient temperatures. If high temperature, material will decompose to Aldehydes, sulphur oxides and oxide of carbon

11. TOXICOLOGICAL INFORMATION

Information on the likely routes of exposure

Inhalation : Prolonged breathing of vapors can cause headache, dizziness, nausea, respiratory irritation or chemical pneumonitis.
Skin contact : Slightly irritating.
Eye contact : Slightly Irritating.
Ingestion : Can cause stomach ache and vomiting.

Symptoms related to the physical, chemical and toxicological characteristics;

Main hazard, if ingested, is aspiration into the lungs and subsequent pneumonitis. Heating can generate vapors that may cause respiratory irritation, nausea and headaches. Inhalation hazard at room temperature is unlikely due to the low volatility of this product.

Delayed and immediate effects and also chronic effects from short and long term exposure; Immediate effects

May cause respiratory irritation, headache, nausea. Mildly irritating to skin and eyes.

Chronic effects:

Prolonged and repeated contact with skin can cause deflating and drying of the skin resulting in skin irritation and dermatitis.

Numerical measures of toxicity

Components	Acute toxicity
Distillates (petroleum), solvent- dewaxed heavy paraffinic	LD50 (oral rat) :> 5000 mg/kg LD50 (Dermal rabbit) :> 5000 mg/kg LC 50 (Inhalation rat) :>2.4 mg/l

Classification of Health Hazards

Acute oral toxicity estimate	Not classified
Acute dermal toxicity estimate	Not classified
Acute inhalation toxicity estimate	Not classified
Skin corrosion / irritation	Category 3
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Safety Data Sheet

HYDRAULIC AW 46



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Respiratory or skin sensitization	not sensitize
Germ cell mutagenicity	Not classified
Carcinogenicity	Highly refined mineral oils are not classified as carcinogenic by the International Agency for Research on Cancer (IARC)
Reproductive toxicity	No data available
Specific target organ toxicity - single exposure	No data available
Specific target organ toxicity - repeated exposure	No data available
Aspiration hazard	No data available

12. ECOLOGICAL INFORMATION

Ecotoxicity

Components	Result
Distillates (petroleum), solvent- dewaxed heavy paraffinic	LC50 Fish (<i>Salmo gairdneri</i>) > 1000 mg/l (96 h) EC50 Crustacea (<i>Daphnia magna</i>) > 1000 mg/l (48 h) ErC50 Algae (<i>Scenedesmus subspicatus</i>) > 1000 mg/l (96 h) NOEC Fish (<i>Pimephales promelas</i>) > 5000 mg/l (7 days) NOEC Crustacea (<i>Ceriodaphnia</i> sp.) 552 mg/l (7 days)

Acute hazards to the aquatic environment estimate: Not classified Long-term hazards to the aquatic environment estimate: Not classified

Persistence and degradability	No data available
Bio accumulative potential	No data available
Mobility in soil	Move under natural forces to the groundwater
Other adverse effects	Long term effect to the aquatic environment.

13. DISPOSAL CONSIDERATIONS

Disposal methods

Waste treatment methods

Disposal as an industrial waste in a manner acceptable to good waste management practice and in compliance with applicable local, state and federal regulations.

Contaminated packaging

Do not attempt to refill or clean containers since residue is difficult to remove. All containers should be disposed of in accordance with governmental regulations.

14. TRANSPORT INFORMATION

UN number	No data available
UN proper shipping name	No data available
Transport hazard class(s)	No regulated

Safety Data Sheet

HYDRAULIC AW 46



PETRONAS

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Packaging group	No data available
Environmental hazards	No data available
Transport in bulk	No data available
Special precautions for user	No data available

15. REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

GLOBAL INVENTORIES

component	USA (TSCA)	EU (EINECS)	AUS (AICS)	JAP (ENCS)	KOR (ELCI)	CHN (IECSC)	PHLP (PICCS)	CAN (DSL/ NDSL)	NZ (NZIOC)
Distillates (petroleum), solvent-dewaxed heavy paraffinic	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Chemical Safety Assessment No data available

16. OTHER INFORMATION

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Full text of other abbreviations

(Q)SAR	-	(Quantitative) Structure Activity Relationship
ACGIH	-	American Conference of Governmental Industrial Hygienists
AICS	-	Australian Inventory of Chemical Substances
ANTT	-	National Agency for Transport by Land of Brazil
ASTM	-	American Society for the Testing of Materials
bw	-	Body weight
CCHC	-	Chemicals Classification and Hazard Communication
CMR	-	Carcinogen, Mutagen or Reproductive Toxicant
CPR	-	Controlled Products Regulations
DIN	-	Standard of the German Institute for Standardisation
DSL	-	Domestic Substances List (Canada)
ECx	-	Concentration associated with x% response
ELx	-	Loading rate associated with x% response
EmS	-	Emergency Schedule
ENCS	-	Existing and New Chemical Substances (Japan)
ErCx	-	Concentration associated with x% growth rate response
ERG	-	Emergency Response Guide
GHS	-	Globally Harmonized System
GLP	-	Good Laboratory Practice
IARC	-	International Agency for Research on Cancer
IATA	-	International Air Transport Association
IBC	-	International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk
IC50	-	Half maximal inhibitory concentration
ICAO	-	International Civil Aviation Organization
ICOP	-	Industry Code of Practice on Chemicals Classification and Hazard Communication
IECSC	-	Inventory of Existing Chemical Substances in China



IMDG	-	International Maritime Dangerous Goods
IMO	-	International Maritime Organization
ISHL	-	Industrial Safety and Health Law (Japan)
ISO	-	International Organisation for Standardization
KECI	-	Korea Existing Chemicals Inventory
LC50	-	Lethal Concentration to 50 % of a test population
LD50	-	Lethal Dose to 50% of a test population (Median Lethal Dose)
MARPOL	-	International Convention for the Prevention of Pollution from Ships
MY PEL	-	Malaysian Permissible Exposure Limit
n.o.s.	-	Not Otherwise Specified
Nch	-	Chilean Norm
NITE	-	National Institute of Technology and Evaluation
NO(A)EC	-	No Observed (Adverse) Effect Concentration
NO(A)EL	-	No Observed (Adverse) Effect Level
NOELR	-	No Observable Effect Loading Rate
NOM	-	Official Mexican Norm
NTP	-	National Toxicology Program
NZIoC	-	New Zealand Inventory of Chemicals
OCSPP	-	Office of Chemical Safety and Pollution Prevention
OECD	-	Organization for Economic Co-operation and Development
PBT	-	Persistent, Bioaccumulative and Toxic
PICCS	-	Philippines Inventory of Chemicals and Chemical Substances
REACH	-	Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals
SADT	-	Self-Accelerating Decomposition Temperature
SDS	-	Safety Data Sheet
STEL	-	Short Term Exposure Limit
TCSI	-	Taiwan Chemical Substance Inventory
TDG	-	Transportation of Dangerous Goods
TSCA	-	Toxic Substances Control Act (United States)
TWA	-	Time Weighted Average
UN	-	United Nations
UNRTDG	-	United Nations Recommendations on the Transport of Dangerous Goods
UVCB	-	Unknown or Variable Composition, Complex Reaction Products and Biological Materials
vPvB	-	Very Persistent and Very Bioaccumulative
WHMIS	-	Workplace Hazardous Materials Information System

Disclaimer

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Product Stewardship Advisory:

PETRONAS aims to increase awareness of all the hazards associated with the storage, handling and use of its products. Thoroughly reviewing the accompanying Safety Data Sheets and disseminating the information to all dependent and interested parties is an essential part of any 'Responsible Care' programme.

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